Embodied AI-Guided Interactive Digital Teachers for Education

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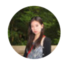

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Embodied AI-Guided Interactive Digital Teachers for Education

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Abstract

Traditional education is considered incapable of providing prompt feedback, facilitating proactive learning, and giving indiscriminate responses. This has been observed in both in-person classes and online courses, especially when students' questions fall outside the instructors' knowledge base or are considered trivial by instructors. Nowadays, the advent of large language models (LLMs) has transformed knowledge acquisition. The LLM-based chatbots enable fast learning through interactive question-answering, which serves as an effective supplement to traditional educational approaches and even shows potential for replacement. To utilize such advancements in education, we propose **MAGI**, a novel system providing E**m**bodied **A**I-**G**uided **I**nteractive digital teachers for education, which integrates LLM-based chatbot technology. To ensure MAGI generates answers without hallucination, we employ a novel retrieval-augmented generation (RAG) paradigm to organize and retrieve useful educational documents for the LLM. Moreover, we create animatable 3D avatars powered by text-to-speech and audio-to-motion models to provide students with interactive conversation experiences. We highlight the possibility of MAGI to enhance education accessibility and improve the overall learning experience.

CCS Concepts

• **Applied computing** → **Education**; • **Computing methodologies** → *Computer graphics*; *Artificial intelligence*.

Keywords

AI for education, large language models, multi-modal generation

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1 Introduction

Education is a fundamental concept that pervades the entire human life. The educational approaches have evolved significantly, from learning through books, and in-person classes, to the current prevalence of online materials and courses. The pursuit of better ways to acquire knowledge is an intrinsic human motivation, reflecting the importance of education in personal development. However, it is observed that these traditional learning methods can not meet the demands for more efficient and effective learning of the general public due to a lack of the ability to provide prompt feedback, facilitate proactive learning, and offer indiscriminate responses [Baker et al. 2022; Nguyen 2022].

In recent years, people have observed the emergence of LLMs, such as GPT4 [Achiam et al. 2023], Llama [Touvron et al. 2023], and Gemini [Team et al. 2023]. They have been used to build LLM-based chatbots to provide human-like interactions for users [OpenAI 2023]. Through interactive conversations powered by LLMs, users can ask questions, find explanations, and obtain prompt feedback, which enables active participation, critical thinking, and a deeper understanding of the knowledge [Dempere et al. 2023; Halaweh 2023]. These posit LLMs' potential to improve the learning experience and facilitate personalized education through AI-driven conversation. However, we observe that most current chatbots are text-based. Although some studies have proposed using digital avatars in education [Anasingaraju et al. 2016; Fink et al. 2024; Johnson and Lester 2016], these avatars have limited knowledge bases and are not powered by LLMs or do not support real-time interactions. Moreover, current LLMs face problems such as domain knowledge gaps, hallucination, and factuality problems [Ji et al. 2023]. Their performance may lag behind task-specific models on knowledge-intensive tasks [Kandpal et al. 2023]. These limitations still pose challenges for LLMs in motivating and improving the learning process.

To address these challenges, we propose MAGI, a system of embodied AI-guided interactive digital teachers providing more accessible and engaging learning experiences. We incorporate the retrieval-augmented generation (RAG) [Lewis et al. 2020] method to augment the LLM with external knowledge to address the hallucination issue. In particular, we develop a new RAG paradigm to organize and retrieve the educational materials for our system. We first collect various educational materials from course slides and online papers as the RAG retrieval database, which are chunked as embeddings and indexed. Then the user input is used to query the database to find the most relevant document, which serves as input for the LLM to generate the final answer. By incorporating documents from different domains, the LLM can be enhanced by various domain knowledge, enabling it to provide more accurate and reliable answers.

Moreover, we build the following pipeline to facilitate real-time interactions between our digital teachers and users. Firstly, given the text response generated by LLM, we employ a Text-to-Speech (TTS) [Pratap et al. 2024] model to convert the text into human-like audio, which serves as the voice for the 3D avatar. Secondly, to provide the avatar with natural face animation that aligns with the audio, we incorporate an audio-driven 3D face animation model [Fan et al. 2022a] to generate the face motion. Specifically, we address the jitter issue by introducing an acceleration loss as a new training constraint. While most 3D face animation models use render methods such as Pyrender, and Blender Python [Fan et al. 2022a; Peng et al. 2023; Thambiraja et al. 2023; Xing et al. 2023], which are unable to realize real-time rendering given long sequences. We improve this problem by utilizing Pytorch3D [Ravi et al. 2020] to accelerate the rendering process, leveraging parallel rendering with GPUs. To further reduce the latency, we propose a simple yet effective implementation of front-end controls to perform rendering while delivering results to users, ensuring seamless video playing. Finally, we improve the visual fidelity of the 3D avatar by incorporating various artist-designed textures and hairstyles.

In summary, our main contributions are:

- We propose MAGI, a system of LLM-based embodied AI-guided interactive digital teachers for education. We design a novel hybrid RAG paradigm to enhance the LLM's ability to provide more accurate and reliable answers based on users' questions.
- To address the latency issue and enable real-time interaction, we utilize Pytorch3D parallel rendering with GPU acceleration and incorporate front-end controls to segment the rendering videos and preload the video chunks sequentially.
- MAGI is a feasible system powered by text-to-speech and audio-to-motion technology that provides realistic and animatable digital teachers.

We envision the interactive digital teachers provided by MAGI as a valuable tool for facilitating learners' engagement, improving learning productivity, and bridging educational gaps.

2 Related Works

2.1 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is a technique to enhance the LLM performance on domain-specific and knowledge-intensive tasks through prompt engineering, which helps bridge the domain knowledge gaps and mitigate hallucination and factuality issues. RAG uses external knowledge resources as a grounding corpus, retrieving relevant documents and passing them to the LLM to improve content generation across various downstream tasks [Jiang et al. 2024; Wang et al. 2024b]. This can be achieved through fine-tuning the LLM or even using a training-free approach [Ram et al. 2023], making it an effective and efficient solution for improving LLM performance. Khandelwal et al. [Khandelwal et al. 2019] first propose using the *k*NN model to extend the language model. Ram et al. [Ram et al. 2023] use rerankers to choose the relevant documents and prepend the retrieved document before the input prefix without further training the language model. Other existing RAG methods [Dai et al. 2022; Li et al. 2023; Zhang et al. 2023] retrieve the prompt-related knowledge based on the embedding similarity, which might cause misaligned or irrelevant chunks to be retrieved. Moreover, some RAG methods [He et al. 2024; Wang et al. 2024a] utilize hierarchical data structures like knowledge graphs to split the knowledge database and search prompt-relevant knowledge efficiently for better retrieving performance. However, these methods also suffer from ambiguity due to the coarse granular data chunks. To address the above problems, we proposed a hybrid RAG method that stores the course content in a hierarchical tree data structure following the nature of the course content structure, enhancing compatibility with educational materials and retrieval accuracy.

2.2 Digital Human Applications

Digital humans are computer-generated representations of virtual human beings capable of performing various tasks in a realistic manner. Digital humans have a vast amount of applications, such as customer service, sales, and news broadcasting. Previous studies have applied digital humans in education, which can be referred to as pedagogical agents. However, these pedagogical agents often lack realism and their knowledge bases are limited to the predefined contents [Anasingaraju et al. 2016; Graesser et al. 2005; Johnson et al. 2011; Johnson and Lester 2016; Rickel and Johnson 1999]. With the rise of AI, digital humans can be more intelligent and realistic. Their knowledge bases can be powered by LLMs and they can be developed using advanced AI technologies to ensure synchronization between speech and lip movements as well as natural expressions and motions. In recent years, many companies introduce their digital humans as commercial products [HeyGen 2021; NVIDIA 2024; UneeQ 2020]. For example, NVIDIA proposes its digital human "James", a digital brand ambassador and virtual assistant to provide accurate product information to customers. However, none of these companies disclose the technical details for achieving realistic avatar reconstruction or real-time interactions. Moreover, most of them are for commercial use, while we believe there is a huge demand in the field of education for digital teachers to help students learn new knowledge more efficiently and effectively. In light of this, our work introduces a practical pipeline for building embodied AI-guided interactive digital teachers for education, enabling real-time interactions.

3 Method

3.1 Overview of the Pipeline

Fig. 2 depicts the pipeline of MAGI. Firstly, we develop a web page to serve as the interface for user interaction with the system. Users input their questions or requirements into a text box on this web page, and these inputs are submitted to the backend. The backend processes these inputs using the RAG technique to retrieve relevant

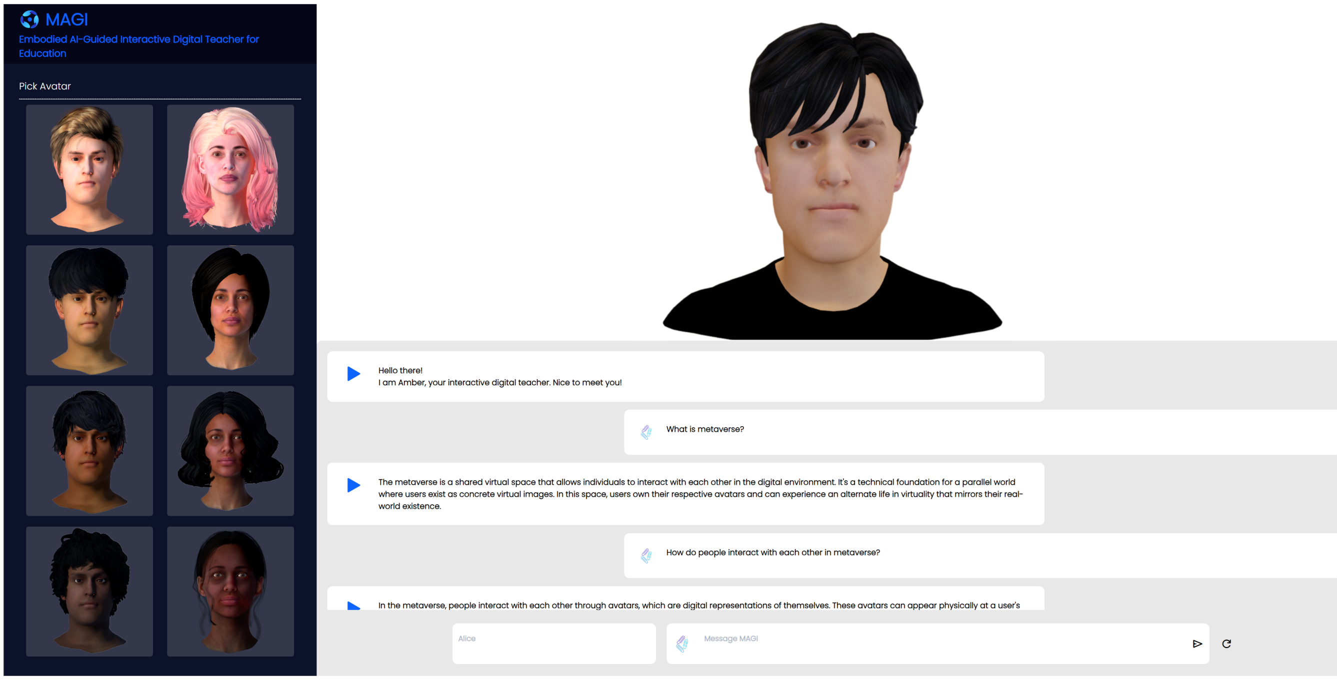

Figure 1: Overview of our deployed system. We propose a system of LLM-based embodied AI-guided interactive digital teachers for education. Our digital teachers can provide personalized education and improve the learning experience for the public.

documents. The retrieved documents are then combined with the input texts and passed to the LLM to generate the final answers to the questions (Stage 1). These final results are converted to audio using the TTS model, which provides the voice for the 3D avatar (Stage 2). We employ an audio-driven 3D face animation generation model to generate the face animation aligning with the audio. Recognizing that realistic and vivid avatars can enhance user experience, we also enable MAGI to customize the textures and hairstyles of its 3D avatars. Once the avatar's animation is rendered, both the rendered video and the final answer are returned to the web page (Stage 3).

3.2 LLM-based Question Answering with Hybrid RAG

LLMs suffer from domain knowledge gaps, hallucination, and factuality issues, which can bring some practical and ethical challenges in educational tasks [Yan et al. 2024]. To address these challenges, we propose a hybrid RAG method that stores the course content in a hierarchical tree data structure following the nature of the course content structure. The hybrid RAG method searches the prompt-relevant knowledge by locating the relevant topics from the hierarchical structure and searching knowledge descriptions with embedding similarity.

The workflow of our hybrid RAG contains three main steps [Gao et al. 2023]: Document Indexing, Document Retrieving, and LLM Prompting.

Document Indexing. In the Document Indexing step, all the documents that contain the ground-truth knowledge of the RAG system are converted into a tree structure based on the course index manually. In this tree structure, each non-leaf node denotes the title or the abstract of a section of the course content and each leaf node denotes the course context stored in the database in the form of a chunk, a compressed text block. The RAG system then recursively generates a short description for each of the non-leaf nodes, which can be seen as the title of a corresponding section based on the contents and the short description of its child nodes using the LLM:

$$\begin{aligned} \mathbf{D}_{tree} &= Tree([\mathbf{d}_0, ..., \mathbf{d}_n]), \\ \mathbf{s}_n &= Summarize(\mathbf{s}_{Children(\mathbf{s}_n)}), \end{aligned} \tag{1}$$

where $\mathbf{s}_i$ denotes the description of non-leaf node n, $\mathbf{d}_i$ denotes the i^{th} document provided, and $\mathbf{D}_{tree}$ denotes the database in tree structure.

For the leaf nodes, their contents are converted into pure texts and split into chunks to accommodate the context limitation of the LLM system:

$$\begin{aligned} \mathbf{T} &= \mathbf{E}_{text}([\mathbf{d}_0, ..., \mathbf{d}_n]), \\ \mathbf{C} &= \mathrm{Split}(\mathbf{T}), \end{aligned} \tag{2}$$

where $\mathbf{d}_i$ denotes the i^{th} document provided, $\mathbf{E}_{text}$ denotes the text extractor for documents in various formats such as image, PDF, and video. $\mathbf{T}$ with length T denotes the total set of the text contents. $\mathbf{C}$ with length L and chunk size c denotes the set of chunks.

The chunks as well as the descriptions of non-leaf nodes will be encoded into semantic embeddings and stored in the vector database:

$$\mathbf{D} = Encoder(\mathbf{C}, \mathbf{S}), \tag{3}$$

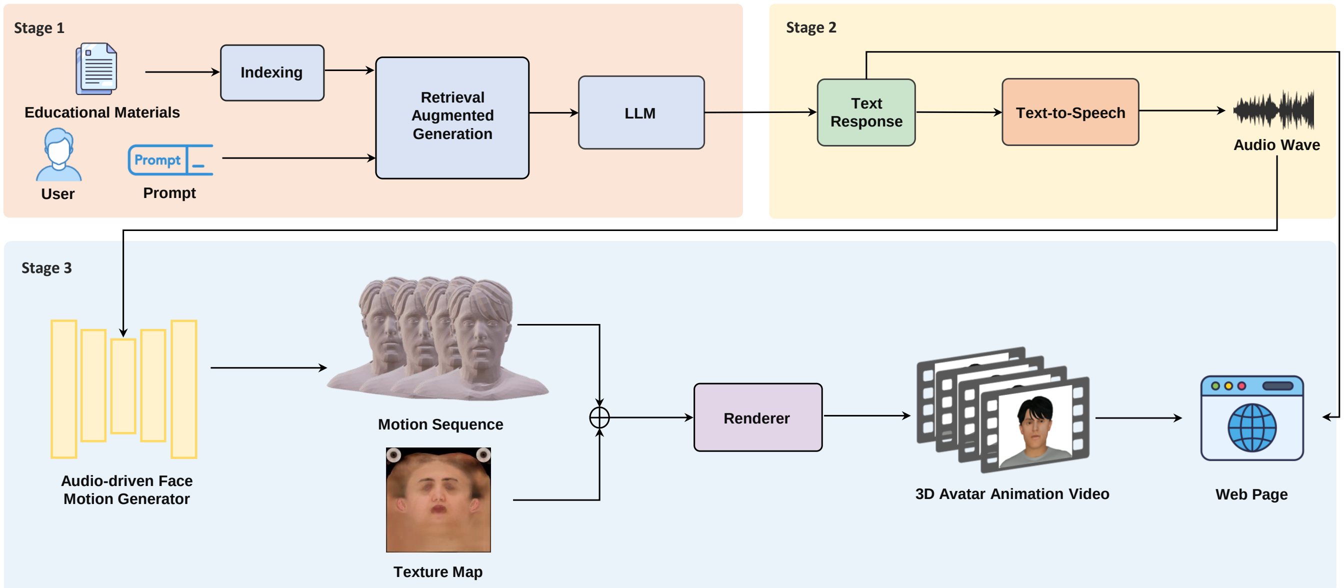

Figure 2: The pipeline of MAGI to create an interactive digital teacher.

where $\mathbf{D} \in \mathbb{R}^{L+N \times d}$ denotes the representative vector database of all chunks and N denotes the number of non-leaf nodes, d denotes the dimension of vectors. Moreover, we use the pre-trained sentence transformers [Reimers and Gurevych 2019] *all-MiniLM-L6-v2* provided by Hugging Face[1] to encode text chunks for a comprehensive semantic extraction performance.

Document Retrieving. In the Document Retrieving step, the RAG system encodes the user input, known as the prompt, and retrieves the most matching knowledge context from the constructed vector database. Notably, our system is a chatbot-based teaching system that provides the service for multiple rounds of conversation. For a prompt entered by the user, all the historical conversations between the user and our system will be treated as additional information. This information will be combined with the prompt for retrieving the most relevant document from the database.

In our hybrid RAG, we first encode the prompt into a vector using the pre-trained sentence transformers:

$$\mathbf{p} = Encoder(\mathbf{prompt}), \tag{4}$$

where $\mathbf{p} \in \mathbb{R}^d$ denotes the vector representation of the prompt. Then, the RAG system will retrieve the non-leaf nodes hierarchically by the vector similarities and collect the contents of the leaf nodes as a candidate knowledge set. With the help of manually describing the contents of each course section and subsections as mentioned before, such tree-search operation can reduce the time-complexity, computational complexity, and accuracy of searching the relevant chunks from the database compared to directly searching the chunks from the whole database. After non-leaf nodes are reached, the RAG system will retrieve the chunks from the candidate knowledge set with the most similar vector representations to the prompt vector:

$$\begin{aligned} \mathbf{CandK}_i &= \mathrm{LeafChild}(\arg\min_{i \in K}(||\mathbf{p}, \mathbf{S}_i||_2)), \\ \mathbf{k} &= \arg\min_{i \in K} ||\mathbf{p}, \mathbf{CandK}_i||_2, \end{aligned} \tag{5}$$

where $\mathbf{S}_i$ denotes the descriptions of i^{th} layer of non-leaf nodes, $\mathbf{CandK} = \{CandK_1, CandK_2, ..., CandK_K\}$ represents the total candidate knowledge set of every layer, which has K candidate chunks in total. $\mathbf{k}$ denotes the index of the retrieved chunks, and the similarity between the prompt vector and the chunk vectors is defined by their L2 distance.

LLM Prompting. For the conversations with historical content, the system first reformulates the latest question referring to historical conversations using the following prompt, with the prompt representation based on [Zhu et al. 2024]. It then retrieves the knowledge context based on the reformulated questions:

```
"System":{
  Given a chat history and the latest user question which might
  reference context in the chat history, formulate a standalone
  question that can be understood without the chat history.
  Do NOT answer the question, just reformulate it if needed and
  otherwise return it as is.
  {Chat History Context}
}
"Human":{Prompt}
```

After retrieving the knowledge context, it will be automatically input into the prompt as a reference for the question answering.

[1] https://huggingface.co/models

The used prompt is as follows:

```
"System":{
    You are an assistant for question-answering tasks.
    Use the following pieces of retrieved context to answer
    the question.
    If you don't know the answer, say that you don't know.
    Use three sentences maximum and keep the answer concise.
    {Retrieved Knowledge Context}
}
```

3.3 Audio-Driven 3D Face Animation

Our audio-driven 3D face animation model follows the architecture of FaceFormer [Fan et al. 2022a], which consists of an audio feature encoder and a face animation generator. To address the jitter problem in the current model, we incorporate an acceleration loss function during the training stage.

3.3.1 Audio Feature Encoder. We employ the state-of-the-art speech model wav2vec 2.0 [Baevski et al. 2020] to extract the audio feature. The wav2vec 2.0 model consists of multiple temporal convolutions layers (TCN) and a transformer encoder [Vaswani et al. 2017]. TCN transforms the raw audio input $\mathcal{X}$ into feature vectors $Z = \{z_1, ..., z_T\}$ for T time steps. A quantization module $Z \longrightarrow Q$ discretizes the audio feature z into speech units q for contrastive training. The transformer encoder $Z \longrightarrow C$ converts the audio feature z into contextualized speech representations c, which is used in multi-modal learning with face motion.

3.3.2 Face Animation Generator. The face animation generator follows the transformer decoder architecture [Vaswani et al. 2017], which consists of multiple self-attention and cross-attention layers. The facial motion sequence $\mathcal{Y}$ is encoded into motion features $F = \{f_1, ..., f_T\}$ by the motion encoder. Along with the positional encodings, the motion features F are projected linearly into queries Q_F, keys K_F of dimension d, and values V_F by the self-attention layers. Temporal bias B_F is added to ensure that the dot product operation is performed over previous frames rather than the future frames:

$$Attention(Q_F, K_F, V_F, B_F) = softmax\left(\frac{Q_F K_F^T}{\sqrt{d}} + B_F\right)V_F, \quad (6)$$

$$B_F = \begin{cases} -Floor((i-j)/p), & j \le i, \\ -\infty, & otherwise, \end{cases} \quad (7)$$

where i and j are indices of B_F, p denotes a specific period, in which indices within the same period have a consistent value, and *Floor* denotes the math floor operation.

Alignment bias B_A is added to the cross-attention module to ensure that the face motion is aligned with the audio:

$$Attention(Q_F, K_A, V_A, B_A) = softmax\left(\frac{Q_F K_A^T}{\sqrt{d}} + B_A\right)V_A, \quad (8)$$

$$B_A = \begin{cases} 0, & ki \le j < k(i+1), \\ -\infty, & otherwise, \end{cases} \quad (9)$$

where the audio tokens A output by the encoder are transformed into keys K_A and values V_A to perform cross-attention with motion queries Q_F. k is a scaling factor to control the alignment range.

3.3.3 Training Objective. We follow [Fan et al. 2022a] to employ a motion loss, which minimizes the ground-truth face motion sequence Y and generator output motion sequence $\hat{Y}$:

$$\mathcal{L}_{motion} = \frac{1}{T}\sum_{t=1}^{T} \|Y^t - \hat{Y}^t\|^2, \quad (10)$$

where Y^t and $\hat{Y}^t$ are the ground-truth face mesh and predicted face mesh at time step t.

To avoid the jitter problem of the generated face motion sequence for unseen audio, we propose employing an acceleration loss. Jitter errors can be represented by acceleration errors and these second-order motion information can benefit the training process to converge faster and better [Zeng et al. 2022]:

$$\mathcal{L}_{acc} = \frac{1}{T-2}\sum_{t=1}^{T-2} \|(Y^{t+2} - 2\times Y^{t+1} + Y^t) - (\hat{Y}^{t+2} - 2\times \hat{Y}^{t+1} + \hat{Y}^t)\|^2. \quad (11)$$

The final training objective is as follows:

$$\mathcal{L} = \mathcal{L}_{motion} + \lambda_{acc}\mathcal{L}_{acc}. \quad (12)$$

3.4 3D Avatar Rendering

To provide users with real-time interaction, we improve the face motion rendering process by leveraging the Pytorch3D renderer [Ravi et al. 2020]. We employ the differentiable neural renderer Pytorch3D to accelerate the rendering process with parallel rendering on GPUs. The textured mesh is rendered into the image domain $\mathcal{I}_t$ by the renderer:

$$\mathcal{I}_t = Renderer(M_t, \tilde{T}, i), \quad (13)$$

where M_t denotes the face mesh of the motion sequence at t time step, $\tilde{T}$ denotes the texture map, and i denotes the illumination in the scene. We batch the 3D meshes in the face motion sequence to render more efficiently.

Furthermore, to enhance the fidelity of the 3D avatar, we design a variety of human textures and hairstyles (see Fig. 3), which include different ages, races, and genders.

3.5 Video Segmentation, Preloading, and Control

To enhance real-time interactions with digital teachers, we design a simple and effective front-end interface that performs video renderings and delivers the rendering results to users. Instead of rendering the entire video at once, we segment the video into parts, each consisting of 60 frames. This setting is based on the rendering speed of Pytorch3D and the video writing speed of FFmpeg. In the front-end interface, we add two *<div>* blocks to show the videos, with the display status of one block set to invisible. Once the two segmented videos are rendered, they are preloaded to the *<div>* blocks and begin delivering to users. Since the next video is preloaded in advance, it ensures seamless video playing like watching a complete video by switching the display status of the two *<div>* blocks. Furthermore, the rest of the videos continue to render while the current video plays for the user.

Figure 3: 3D avatars in different hairstyles and textures.

Table 1: Comparison of lip vertex error (LVE) and acceleration error (Accel) on VOCA test set with and without acceleration loss.

Training Objective	LVE (mm) ↓	Accel ($mm/frame^2$) ↓
w/o $\mathcal{L}_{acc}$	5.43	8.89
w $\mathcal{L}_{acc}$	**5.31**	**8.23**

4 Results

4.1 Implementation Details

For the LLM backbone, we use the Llama 3 8B [Dubey et al. 2024] and deploy it in 4 NVIDIA A5000 GPUs. We train the audio-driven face motion generator for 100 epochs. For the training loss weighting factors, we set λ_{acc} to 0.1. The rendering batch size is 16.

4.2 Visualization of the System

The visualization of MAGI is shown in Fig. 1. The system interface is divided into three sections: the lower section displays the conversation and the text input box, the upper section presents the digital teacher, and the left section displays different digital teachers.

4.3 Ablation Study

4.3.1 Effect of Acceleration Loss. We examine the effect of incorporating an acceleration loss in the training objective of the audio-driven face animation model. Lip vertex error (LVE) is a widely used lip-sync metric for measuring lip movement quality [Fan et al. 2022b; Peng et al. 2023; Richard et al. 2021], which calculates the average $l2$ error of the lips in the test set. We use Accel to measure smooth prediction following [Choi et al. 2021; Kanazawa et al. 2019; Zeng et al. 2022], which calculates the acceleration error between ground truth 3D acceleration and predicted 3D acceleration of each lip vertex. As shown in Tab. 1, the model can achieve lower lip vertex error and acceleration error on the VOCA test set [Cudeiro et al. 2019] by incorporating an acceleration loss. We visualize the acceleration error on one of the identities in the VOCA test set. As shown in Fig. 4, adding acceleration loss contributes to smaller and smoother errors.

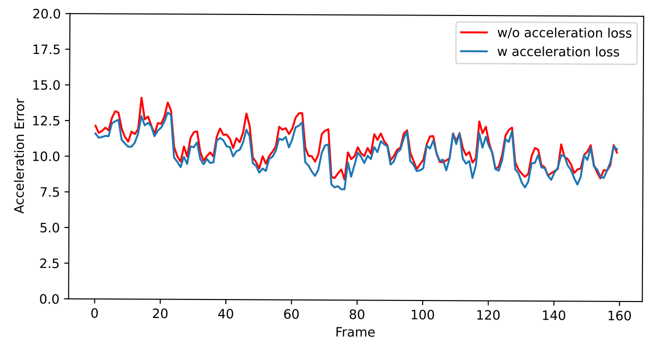

Figure 4: Comparison of acceleration error on VOCA test set. Training without acceleration loss contributes to acceleration error spikes compared to training with acceleration loss.

Table 2: Comparison of rendering times for different render methods on 3k frames.

Render Method	Time ($second$) ↓
Pyrender	652
Blender Python	290
Pytorch3D	33

4.3.2 Rendering Speed Comparison. We compare the rendering speed of three different methods, Pyrender, Blender Python, and Pytorch3D, for a long audio clip (108 seconds) at a frame rate of 30 FPS, resulting in nearly 3k frames. As shown in Tab. 2, using Pytorch3D as the renderer significantly accelerates the rendering process for a long video compared to the other methods.

4.4 Example Interactions with Digital Teacher

Fig. 5 shows some examples of the conversations with the digital teacher in our system. We use metaverse education as a use case. These conversations demonstrate the system's capability to provide accurate answers regarding metaverse knowledge and show the effectiveness of our RAG method.

5 Conclusion and Future Work

In this paper, we propose MAGI, a system of LLM-based embodied AI-guided interactive digital teachers for education. We design a new RAG paradigm to extend the LLM with educational materials, including various course slides and online papers, enabling our chatbot to provide more accurate and reliable responses. Moreover, we use parallel rendering with GPU acceleration and incorporate front-end controls to realize real-time interactions. Additionally, we incorporate 3D avatars acting as digital teachers in the system to enhance the user experience during conversations. We believe these interactive digital teachers have the potential to improve education accessibility and improve the learning experience for the general public, thereby providing high-quality learning opportunities regardless of a student's location or background.

There are several avenues for future work. While we have successfully implemented the technical pipeline, our UI design can be further improved to be more user-friendly. We plan to conduct

Figure 5: Examples of conversation with our chatbot on metaverse knowledge.

user studies to collect feedback from users to help us improve the system. Furthermore, we aim to support more languages beyond English leveraging AI technology, making the system accessible to audiences around the world. We also plan to support diverse types of interactions, such as voice commands, gestures, and augmented reality interfaces, to provide a richer and more engaging user experience.

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